

Problem of the Week

Due Monday, October 5 at 11:59 pm (Arizona time)

One in Front

Find all integers $n \geq 1$ with the property that writing the digit 1 in front of the decimal representation of n^2 produces exactly n^3 . For instance, if n^2 were 37, the new number would be 137.

How to enter

Scan the code, or go to ac.nau.edu/lms-apps/self-enroll/545026, to join the POTW Canvas course. Then upload **one** PDF (typed, or neatly handwritten and scanned) to this week's assignment. Open to all NAU undergraduates.

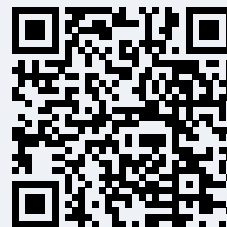
The rules

- **Your own work only.** No AI tools (ChatGPT, Gemini, Claude, Copilot, Photomath, ...), no internet, no calculators or software, and no help from other people.
- **Explain everything.** A bare answer earns little credit; a clear, complete argument earns full credit.
- Any participant may be asked to explain their solution in person.

Scored 0–4. Full rules, past problems, and model solutions:

naumathstat.github.io/problem-of-the-week

Questions: ian.williams@nau.edu



Join on Canvas